

Kexin Wei

+65 89206264 | weikexin611@gmail.com | kexin-wei-research | kexinwei.org

AI, robotics, and cybersecurity engineer with expertise in full-stack development and team leadership.
Strong focus on system architecture, quality control, and cross-functional collaboration.

Experience

Research Engineer

Mar 2025 — Present

Imperial Global Singapore, Imperial College London.

Singapore

- **AI Agents:** Building an agentic (Claude tool-use, multi-step planning) penetration-testing system for medical cyber-physical systems.
- **Simulation:** Translated the MATLAB controller to Python and tuned a real-time closed-loop glucose-insulin patient simulator.
- **Full-Stack:** Built a Flask + Vue.js application for real-time simulation and cyberattack visualization.
- **DevOps:** Dockerized and deployed the simulation stack with multi-container orchestration.
- **Infrastructure:** Administers lab workstations, servers, and container infrastructure.

Software Team Lead

Mar 2024 — Jan 2025

Creative Medtech Solutions Pte.Ltd.

Singapore

- **Robot:** Engineered robotics control systems including kinematics optimization for 4-DoF robot and control interfaces.
- **AI:** Implemented AI-based medical image segmentation, 4D DICOM processing, and US/MRI image registration algorithm.
- **CyberSecurity:** Enhanced security through encryption protocols, SQL injection protection, and streamlined Qt based GUI.
- **Lead:** Led a team of 6 engineers: Git workflow, Scrum adoption, GoogleTest integration, and code-review standards.

Research and Development Engineer

Jun 2022 — Mar 2024

Creative Medtech Solutions Pte.Ltd.

Singapore

- **Robot:** Designed a motion-tracking algorithm for a 6-DoF robot arm integrating an A*STAR CNN, verified in vivo across pig and dog trials.
- **AI:** Developed and validated an AI-based 3D medical image segmentation pipeline for prostate biopsy guidance.
- **Image:** Prototyped kidney segmentation solutions using SAM, MedSAM, and OpenCV for potential implementation.
- **Sensor:** Integrated electromagnetic sensor system in the HIFU treatment software for 3D visualization of the ultrasound probe.
- **Signal Processing:** Created HIFU power-calibration and kidney respiratory-tracking algorithms, verified in ex-vivo and animal trials.
- **Patent:** Filed 3 patents in medical algorithm design: HIFU calibration, respiratory tracking, and time synchronization.

Research Assistant — Internship

Jul 2018 — Sep 2018

Beijing Surgerii Technology Co., Ltd. and Kii Lab of SJTU under Professor Kai Xu

Shanghai, China

- Conducted 20+ experiments to evaluate the three-dimensional positioning algorithm of a multi-backbone robot.
- Assembled and tested the motor-driven part of a surgical robot arm to ensure fundamental functions.
- Analyzed needle insertion experiment results and redesigned mechanical structure using AutoCAD for improved performance.

Education

M.Eng in Medical Robotics and AI

Aug 2020 — May 2022

National University of Singapore, Singapore

GPA: 4.63/5.00

- Thesis: Deep Reinforcement Learning (DRL) for Robot-assisted Surgical Training. [Code](#).
- TA for ME2143E Motor Characteristics and ME5405 Machine Vision.

B.Eng in Mechanical Engineering

Sep 2014 — Jul 2019

Tongji University, China

GPA: 4.61/5.00

Publications & Patents

- **K. Wei**, I. Zahid, L. Cheng, A. Bird, V. Schlegel, and N. Oliver, “Generalisability of the Glucoprint: CGM-Based Re-identification Across Six Type 1 Diabetes Cohorts,” Imperial College London / Imperial Global Singapore. *In preparation*.
- L. Cheng, **K. Wei**, V. Schlegel, Q. Shao, A. Bird, W. Wei, N. Oliver, and A.A. Bharath, “G-FM: Imputation-Guided Flow Matching for Multivariate Time-Series Generation,” Imperial College London / Imperial Global Singapore. *Submitted*.
- Y. Niu, Y. Ma, S. Nandakumar, M. Chen, V. Schlegel, **K. Wei**, L. Cheng, A. Bird, A.A. Bharath, and S.K. Lam, “HealthLoopQA: A Context-Aware Question Answering Benchmark for Interpreting Wearable Monitoring Data in Diabetes Care,” Imperial Global Singapore / Imperial College London / NTU. *Submitted*.
- A.R. Bird*, V. Mohan*, **K. Wei**, Z. Cao, S.H. Ng, M.M. Mulay, V. Schlegel, M.G. Moffett, J.E. Moore Jr, C.-H. Chang, N. Oliver, and A.A. Bharath, “Toward Trustworthy Closed-Loop Diabetes Technologies: A Dynamic Platform for Safety and Security Evaluation,” Imperial Global Singapore / Imperial College London / NTU. *Submitted*.
- **3 Patents Filed**: HIFU power calibration algorithm, robot respiratory tracking algorithm, and HIFU sweep mode time synchronization.

Awards & Honors

AI	Hackathon “Runner Up” Prize at bio LLMs hosted by 4Catalyzer (2023)
Mathematics Modeling	2nd Prize in CUMCM (2018); Honorable Mention in MCM/ICM (2018)
Scholarship	3rd Class Award (2018); 1st Class Award (2016); 3rd Class Award (2015)

Skills

AI/ML	PyTorch, TensorFlow, Deep Learning (CV, CNN, DRL), LLM (Claude, OpenAI, Gemini; LangChain/LangGraph)
Robotics	ROS, 6-DoF motion control, kinematics, EM/US tracking, DICOM
Programming	Python, C++, MATLAB, SQL, Rust
Frameworks	Qt, Flask, Vue.js, Flutter, OpenCV, VTK
Tools & Infra	Docker, Git, Linux (Ubuntu/RHEL), CI/CD
CAD/CAE	Solidworks, AutoCAD
Language	English (C1), German (C1), Mandarin (native)

Community & Service

- **Open source**: Contributor to Women Devs SG projects — the community website (merged) and the *bibsnub* childcare-finder app.
- **Junior Developers SG**: Participant in coding dojos — test-driven development, refactoring, and test automation.
- **Division Zero (div0)**: Cybersecurity workshops — home-lab building, SHELLgym, and Hack Your First Flag.